



---

**HK IMO Training 2019-2020**  
**Problem Set 2**  
**For 5 October 2019**

---



**Problem 5.** Let  $ABC$  be a triangle. Let  $\ell$  be the line through  $B$  perpendicular to  $AB$ . The perpendicular from  $A$  to  $BC$  meets  $\ell$  at the point  $D$ . The perpendicular bisector of  $BC$  meets  $\ell$  at the point  $P$ . Let  $E$  be the foot of the perpendicular from  $D$  to  $AC$ . Prove that triangle  $BPE$  is isosceles.

**Problem 6.** (a) Given are 2019 pairwise distinct integers having no odd prime divisors smaller than 37. Prove that there are two of these integers whose sum has no odd prime divisors smaller than 37 either.

(b) Does the statement remain correct if both occurrences of number 37 are replaced with 38?

**Problem 7.** In the country of Oddland, there are stamps with values 1 cent, 3 cent, 5 cent, etc, one type for each odd number. The rules of Oddland Postal Services stipulate the following: for any two distinct values, the number of stamps of the higher value on an envelope must never exceed the number of stamps of the lower value.

In the country of Squareland, on the other hand, there are stamps with values 1 cent, 4 cent, 9 cent, etc, one type for each square number. Stamps can be combined in all possible ways in Squareland without additional rules.

Prove for every positive integer  $n$ : In Oddland and Squareland there are equally many ways to correctly place stamps of a total value of  $n$  cent on an envelope. Rearranging the stamps on an envelope makes no difference.

**Problem 8.** Find all functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfying the equality

$$f(f(x) + f(y)) = (x + y)f(x + y) \quad \text{for all } x, y \in \mathbb{R}.$$